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SemanticDistance: An R package for Computing Semantic Distance in Structured Texts and Visualizing Clustering Properties in Unstructured Word Lists

Jamie Reilly[1, 2], **Emily B. Myers**[3, 4], **Hannah Mechtenberg**[4], and **Jonathan E. Peelle**[5, 6, 7]

1 Department of Communication Sciences and Disorders, Temple University, United States **2** Department of Psychology and Neuroscience, Temple University, United States **3** Department of Speech, Language, and Hearing Sciences, University of Connecticut, United States **4** Department of Psychological Sciences, University of Connecticut, United States **5** Institute for Cognitive and Brain Health, Northeastern University, United States **6** Department of Communication Sciences and Disorders, Northeastern University, United States **7** Department of Psychology, Northeastern University, United States

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Software

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Summary

`SemanticDistance` computes pairwise distance between units of language (e.g., word-to-word, ngram-to-word, turn-to-turn) in both structured language samples and unstructured word lists. `SemanticDistance` has cleaning and formatting options including stopword removal and lemmatization. The package computes two complementary cosine distance indices for each pairwise contrast of interest. `SemanticDistance` can also be used to examine clustering properties within unstructured word lists, generating dendrograms and simple igraph network plots.

Software Design

`SemanticDistance` was is an open-source R package designed to accommodate a wide range of user expertise (e.g., neuroscience, linguistics, computer science). The software does not leverage artificial intelligence or use any proprietary dependencies. The package instead indexes an internal lookup database containing publicly available lexical norms.

Statment of need

`SemanticDistance` is unique in its capacity to compute semantic distance metrics within running language samples (e.g., word-to-word, bigram-to-word) and to produce semantic network visualizations. The package also provides complementary (but different) semantic distance metrics (experiential vs. embedding) computed over the same stimuli. This package will support novel insights into the flow of meaning within naturalistic language samples.

Research Impact Statement

SemanticDistance has supported one peer-reviewed publication to date in the journal, Cortex (Mechtenberg et al. 2026) along with one refereed conference presentation at the annual conference of the Society for the Neurobiology of Language. Importantly, the application enables a variety of analyses that have been formerly difficult or impossible, including comparing semantic distance measures with other measure of prediction in language, including those generated by large language models.

Description

Semantic distance has been broadly defined an empirical measure of conceptual relatedness between two or more words situated within an n-dimensional space (Reilly et al. 2025). Semantic spaces differ in their dimensionality and biological plausibility. As an R package, `SemanticDistance` bundles text cleaning and formatting options (e.g., stopword removal, lemmatization) with two two complementary semantic distance metrics that characterize/quantify similarity between pairs of language constituents (e.g., words, ngrams, turns). `CosDist_Glo` reflects cosine distance between vectors derived from training a GLOVE word embedding model (Pennington, Socher, and Manning 2014). `CosDist_SD15` refects cosine distances derived from a 15 dimension sensorimotor and affective space derived from explicit human ratings (Reilly et al. 2023).

`SemanticDistance` computes indices of relatedness between lexical constituents within the following text formats: 1: **monologues**: stories, narratives, and structured text
2: **dialogues**: turn-to-turn within two-person conversation transcripts.
3: **word pairs in columns**: word pairs
4: **unordered lists**: bags-of-words

One of the most innovative functions of `SemanticDistance` is its capacity to generate rolling measures of semantic distance within stories or narratives with chunk sizes (e.g., word-to-word, turn-to-turn). Chunk size is an optional argument for many of the software's functions. For example, if a user specified a chunk or n-gram size of 5 words, the software would automatically compute a rollinhg measure of semantic distance betweem each new word in a language sample relative to the five words that preceded it.

AI Usage

`SemanticDistance` uses an internal database composed of many words with a fixed set of embeddings. The software does not use AI or rely on proprietary package dependencies. We did use GPT 4.0 (OpenAI) to troubleshoot R coding errors and assist with generating complex regular expressions (regex) used for cleaning text data.

References

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